12-08 第1篇

阅读理解（说明文） Why Do Cats Almost Always Land on Their Feet?

① If you have ever watched a house cat slip from the edge of a sofa — only to twist sharply in the air and land, a tenth of a second later, softly on its four paws — you have seen one of biology's quietest small miracles. This ability has a careful name in science: **the cat righting reflex**. It was photographed for the first time by the French physiologist **Étienne-Jules Marey in 1894**, using an early high-speed camera (60 frames per second) that captured a falling cat's whole twist in only **0.4 seconds**. What the pictures showed — a cat going from back-down to paws-down without, apparently, pushing against anything in mid-air — was so strange that several scientists at the time quietly suspected the photographs were fake. They were not. The cat, it turns out, is a small soft gymnast using three quiet tricks of physics all at once.

② **Trick one: a spine that can split in half.** A cat's backbone is unusually flexible. It has **53 vertebrae** (humans have 33), and its lower back can rotate almost **180°** independently from its upper back. Imagine twisting your shoulders one way and your hips the other way, at the same time — most of us cannot do this at all. A cat can. When a cat falls, it first determines which way is "up" through its **inner-ear balance organ**, the vestibular system, in about **0.08 seconds**. Then, within another **0.15 seconds**, its front half and back half begin rotating in different directions.

③ **Trick two: pulling in and pushing out the legs.** This is the clever physics. Imagine a spinning skater: when she **pulls her arms in**, she spins faster; when she **stretches them out**, she spins slower. This is **conservation of angular momentum** — a core rule of physics. A cat, in the air, uses this rule twice. It first **pulls its front legs in** while its back legs are stretched out; this lets the **front half rotate quickly** to face down. Then it **stretches the front legs out and pulls the back legs in**; now the **back half rotates** into alignment. Net result: the cat has turned itself over, in mid-air, **without pushing on anything**. The total turn takes about **0.3 seconds**.

④ **Trick three: a tail as a small rudder.** A cat's tail, while not the main instrument, acts as a **fine-adjustment rudder**. A 2022 study at Georgia Institute of Technology filmed 47 falling cats with 400-frames-per-second cameras and showed that tails contribute roughly **12%** of the total righting-rotation in shorter falls. Cats without tails (such as Manx cats, born tailless) can still right themselves, but their falls are less smooth and their landings slightly rougher. In very short falls — less than the length of a bookshelf — the tail's role is more important, since there is little time to rely on the full spine-twist.

⑤ **Minimum fall height.** The righting reflex has a well-measured lower limit: a cat needs at least **30 cm of falling distance**, plus about **0.4 seconds in the air**, to complete the full twist. Below this, the cat may land awkwardly on its side. A famous 1987 study by the New York Animal Medical Center analysed **132 cats that had fallen from apartment windows**. The surprising result, since confirmed many times: fall injuries were, on average, **worst from the 5th to the 7th floor** (about 10—20 m), but **less severe from above the 7th floor**. Why? Cats falling from higher floors had more time to fully spread their legs and flatten like a "parachute", reducing terminal velocity to about **95 km/h** (humans reach **180—200 km/h**); the slower hit spread impact across four paws and a padded belly, not a single bone.

⑥ **Why did evolution give cats this reflex?** Cats are **ambush hunters in trees**. For 30 million years, their ancestors climbed branches, balanced on narrow twigs, and sometimes fell. A cat that could twist and land on its paws was, on average, far more likely to survive the fall — and go on to hunt again. Over many generations, the trait became nearly universal in the cat family, from domestic cats to leopards and tigers. (Tigers, too, can right themselves, although weighing 200 kg makes the landing much less graceful.) By contrast, **dogs**, evolved mostly as ground-running hunters, **do not have the reflex** — a dog dropped from a short height will mostly fall sideways.

⑦ **What about kittens?** Newborn kittens do **not** have the reflex; it develops between about **3.5 and 7 weeks old**. A kitten dropped gently onto a cushion at 4 weeks may flop onto its side; by 7 weeks, the full twist works. Veterinarians today recommend that nobody deliberately test this on a kitten — the risks are real and the experiment is, in any case, answered.

⑧ **A small safety note.** The cat righting reflex is beautiful but not magic. A cat falling from a table still needs the 0.4 seconds and at least 30 cm; a cat flung or caught mid-air can still be hurt. In Chinese cities, where 95% of households live above the ground floor, **cat-falling injuries rise every summer**, when windows are opened for cooling. A small rule: **install window screens in flats with cats**. A cat is a small gymnast; we are the room managers.

⑨ So, the next time a cat balances on the edge of your balcony and looks, half-interested, at the sky, remember what is inside her: a **flexible spine**, a **fine-tuned inner ear**, **legs that pull and push like a spinning skater's arms**, and a **tail that quietly adjusts**. For thirty million years, that small package has been quietly improved. It is one of the reasons that, in the world of small animals, **the soft landing of a cat** is one of the most reliable events in nature.

注释： reflex 反射；vertebra（复数 vertebrae）椎骨；angular momentum 角动量；rudder 舵；terminal velocity 终端速度；ambush 伏击

根据短文内容，选择最佳答案。

1. Who first photographed the cat righting reflex, and what did the pictures show?

A. Isaac Newton, in 1680, showed a cat falling from a tree.
B. Étienne-Jules Marey, a French physiologist, in 1894, using a 60-frame-per-second early high-speed camera captured a falling cat's whole twist in only 0.4 seconds; the pictures showed a cat going from back-down to paws-down without, apparently, pushing against anything in mid-air.
C. Thomas Edison, in 1920, filmed a kitten falling.
D. An unknown scientist in 1700.

2. How does a cat use "conservation of angular momentum" during a fall (Paragraph ③)?

A. By pushing hard against the air.
B. By pulling its front legs in while back legs are stretched out (front half rotates quickly to face down), then stretching the front legs out and pulling the back legs in (back half rotates into alignment); the whole turn takes about 0.3 seconds and requires no pushing against anything.
C. By flapping its ears.
D. By swallowing air.

3. What did the 1987 study by the New York Animal Medical Center of 132 cats falling from apartment windows find?

A. Cats always die at high floors.
B. Injuries were on average worst from the 5th to 7th floors (about 10—20 m) but less severe from above the 7th floor, because cats falling from higher floors had more time to spread their legs and flatten like a "parachute," reducing terminal velocity to about 95 km/h (humans reach 180—200 km/h); the slower impact was spread across four paws and a padded belly.
C. Injuries were equally bad at all heights.
D. Cats bounce off walls.

4. According to Paragraph ⑦, when does the righting reflex develop in kittens?

A. Before birth.
B. At 1 day old.
C. Between about 3.5 and 7 weeks old; a kitten dropped onto a cushion at 4 weeks may flop sideways, but by 7 weeks the full twist works. (Vets recommend not testing this on kittens.)
D. At six months old.

5. What is the main idea of the final paragraph?

A. Cats are magical creatures.
B. The cat's soft landing is one of the most reliable events in nature because inside her

is a flexible spine, a fine-tuned inner ear, legs that pull and push like a spinning skater's arms, and a tail that quietly adjusts — a small biological package that has been improved by thirty million years of evolution.
C. Cats are better than dogs.
D. Cats can fly short distances.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第①段 1894 Marey
【解析】"1894 / 60 fps / 0.4 秒"直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段角动量
【解析】"前收后伸 → 前伸后收 / 0.3 秒 / 无推点"直引。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第⑤段 1987 纽约研究
【解析】"5-7 楼最重 / 7 楼以上更轻 / 95 km/h"直引。A、B、D 均错。故选 C。

4.
【答案】C
【题型】细节理解题
【定位】第⑦段幼猫
【解析】"3.5-7 周"直引。A、B、D 均错。故选 C。

5.
【答案】B
【题型】主旨大意题
【定位】第⑨段
【解析】"四件 + 3000 万年演化"主旨。A、C、D 均偏离。故选 B。